Dr. Alex Skeels

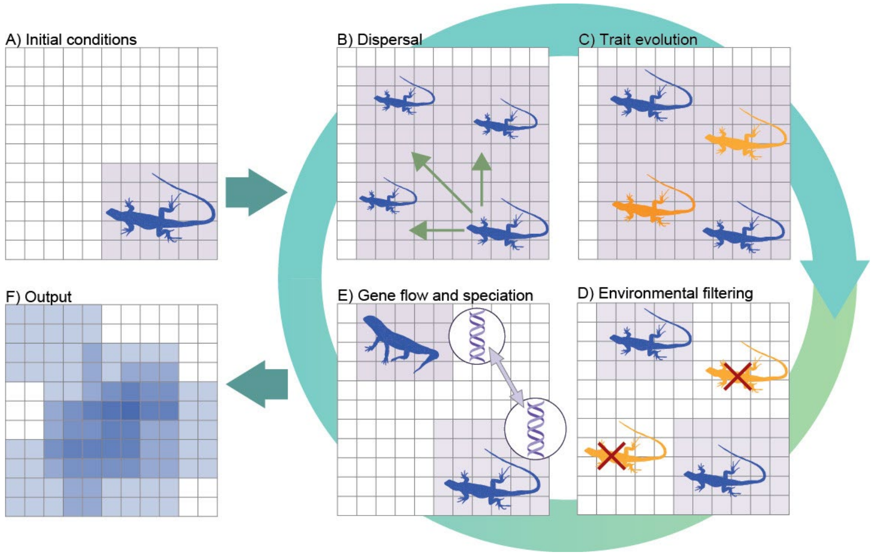
Week 7:
Species in Space I

An introduction to Species Distribution Modelling

BIOL3207 – Data Science for Biologists

About me

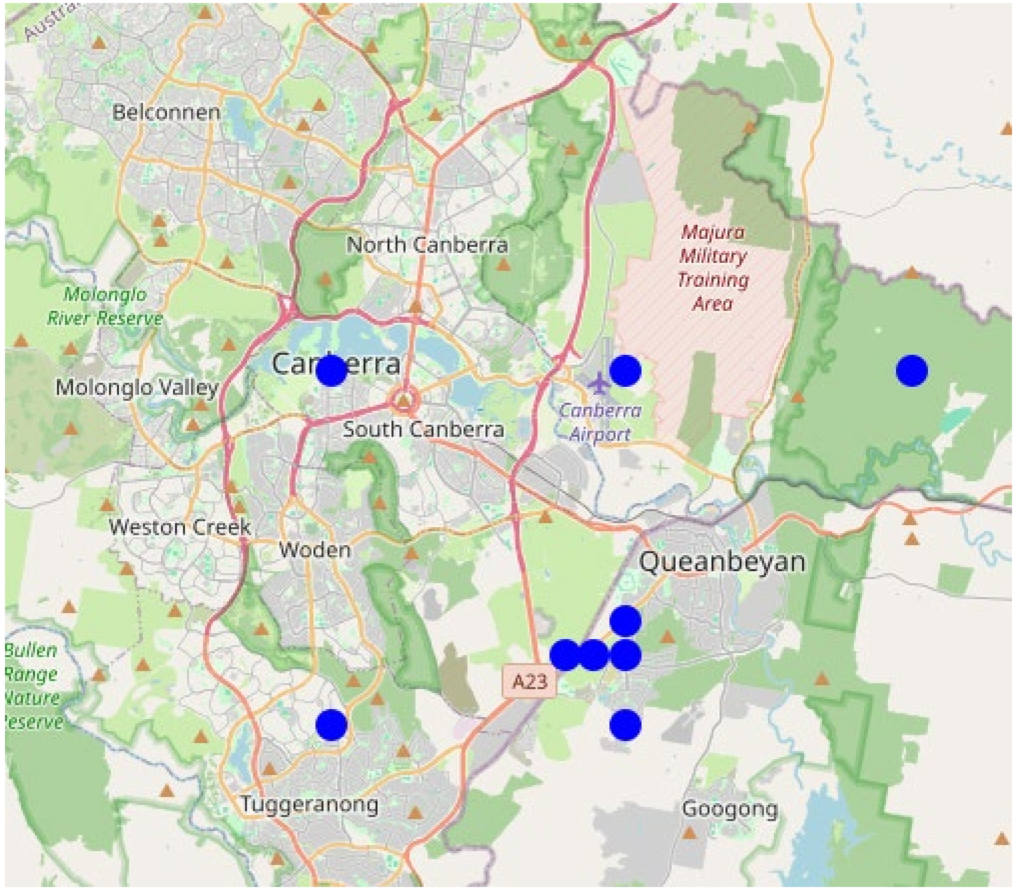
- Research Fellow at ANU (2025-onwards)
  - how environmental changes shapes the origin and decline of biodiversity across the Asia-Pacific region
- Biogeography / Macroevolution / Macroecology
- Before this:
  - Postdoctoral researcher at ANU (2023-2025) and ETH Zurich (2020-2023)
  - PhD at ANU with Marcel Cardillo (2016-2020)
- I work with spatial and phylogenetic data and use simulations and statistical models to test hypotheses about the evolution of biodiversity



Weeks 7 and 8

Date	Session	Subject
Week 7	Lecture 1	Species in Space I: An introduction to species distribution modelling
Week 7	Workshop 1	Points, polygons, and geometric SDMs in R
Week 8	Lecture 2	Species in Space II: More species distribution modelling
Week 8	Workshop 2	Climate data and statistical SDMs in R

Where is this species distributed?



Canberra Grassland Earless Dragon
Tympanocryptis lineata

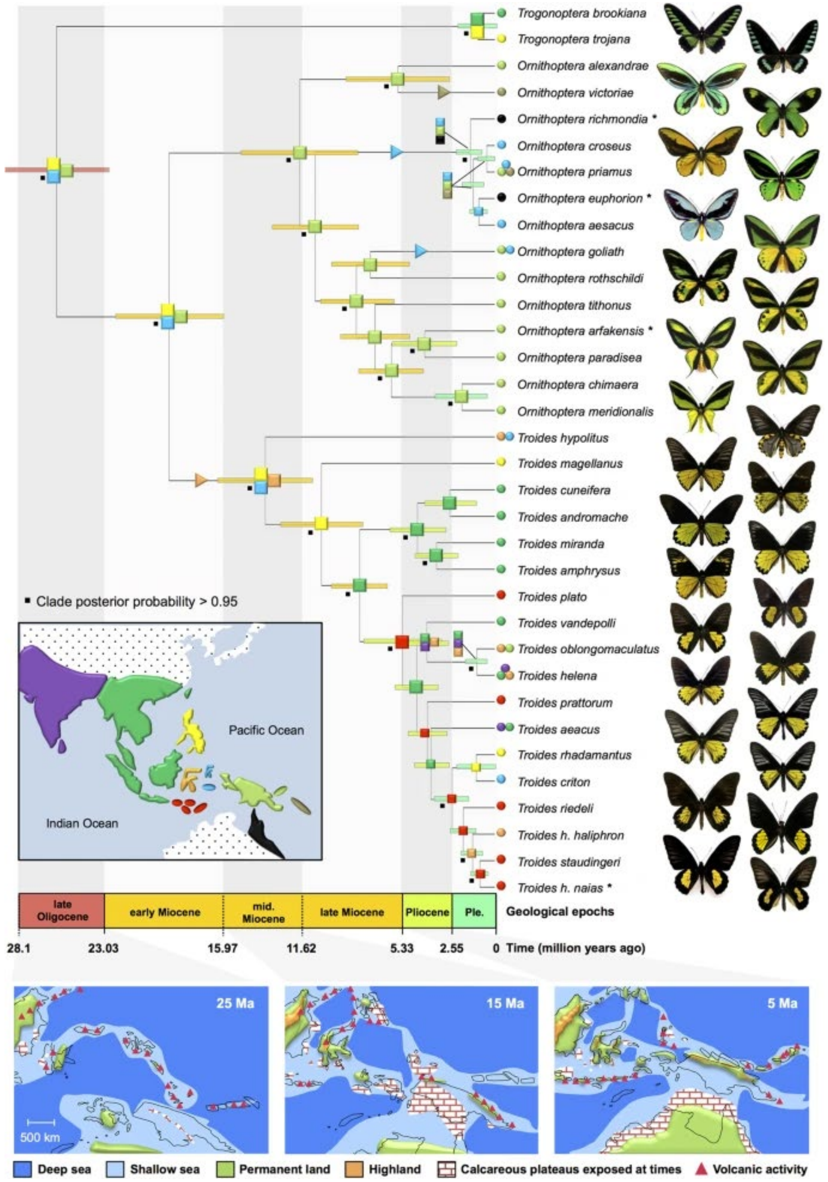
Species Distributions are Limited

- No species is found everywhere
- Distributions are determined by complex biological and environmental factors
  - Evolutionary history - where did the species originate?
  - Environmental preferences - what habitats does it require?
  - Ecology - what does it need to live and reproduce?
  - Dispersal abilities - where can it get to?
- *Biogeography* aims to understand these factors


Alfred Russell Wallace

Applications

- ▸ **Evolutionary Biology**
- Conservation
- Invasive Species Management
- Climate Change Impacts
- Disease Management



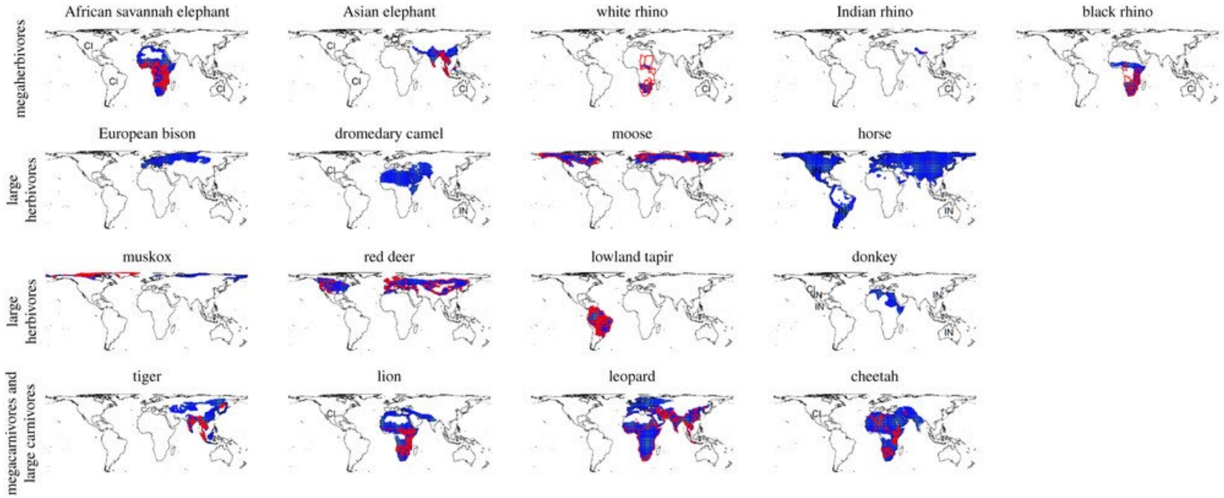
Condamine et al. 2015. Sci. Rep

Applications

- Evolutionary Biology
- ▸ **Conservation**
- Invasive Species Management
- Climate Change Impacts
- Disease Management



Rewilding – reintroducing megafauna species to former range

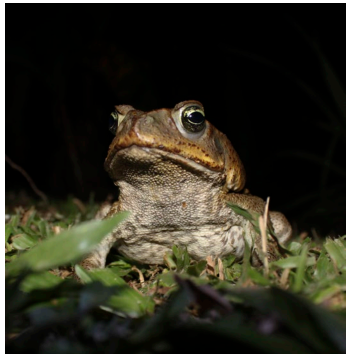


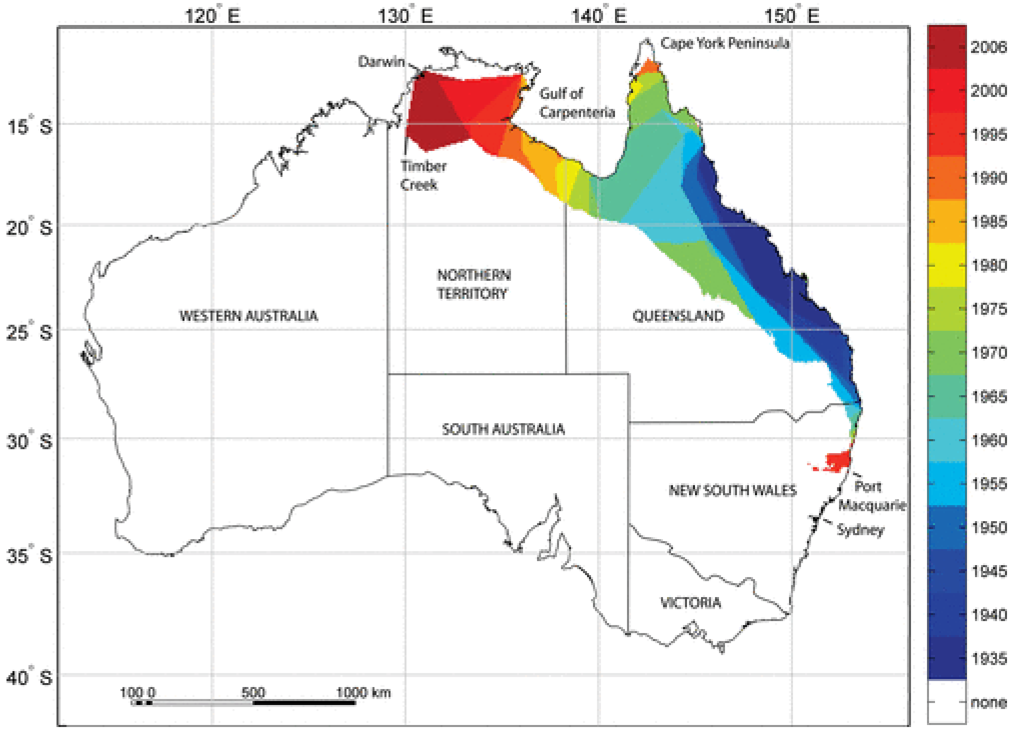
Jarvie and Svenning. 2018. Phil. Trans. R. Soc.

Applications

- Evolutionary Biology
- Conservation
- ▸ **Invasive Species Management**
- Climate Change Impacts
- Disease Management

Monitoring the spread of Cane Toads in Australia



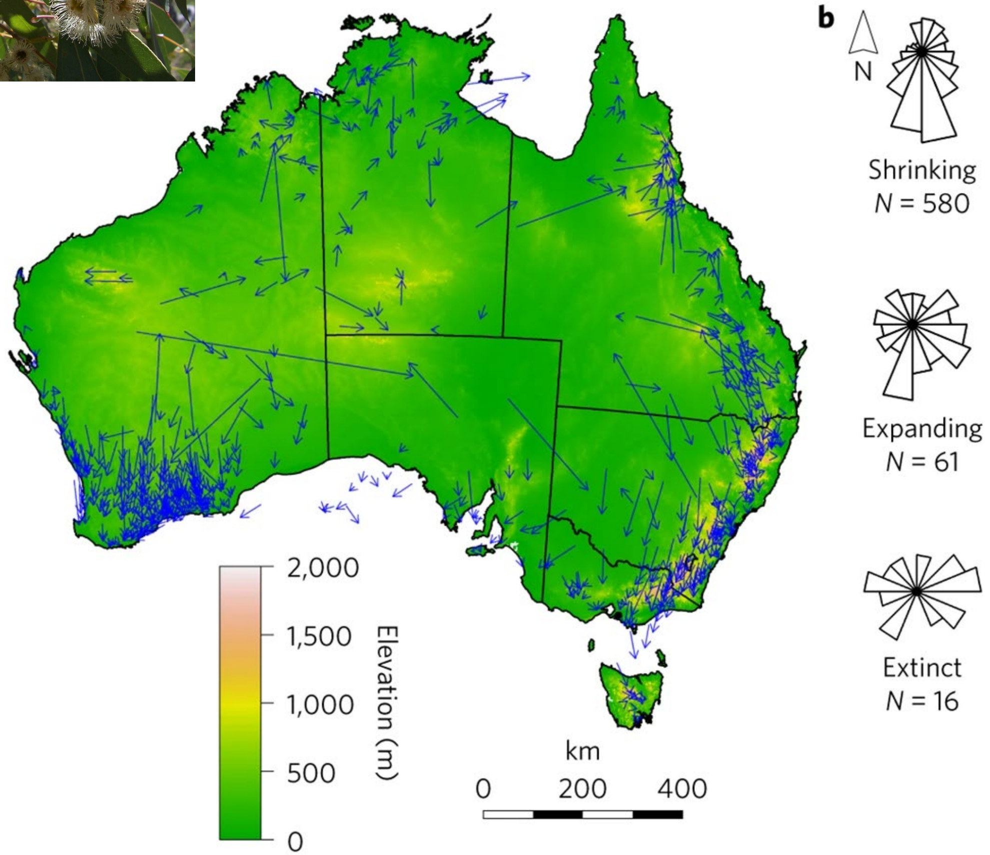


Urban et al. 2008 American Naturalist

Applications

- Evolutionary Biology
- Conservation
- Invasive Species Management
- ▸ **Climate Change Impacts**
- Disease Management

Range shifts under climate change in Eucalyptus

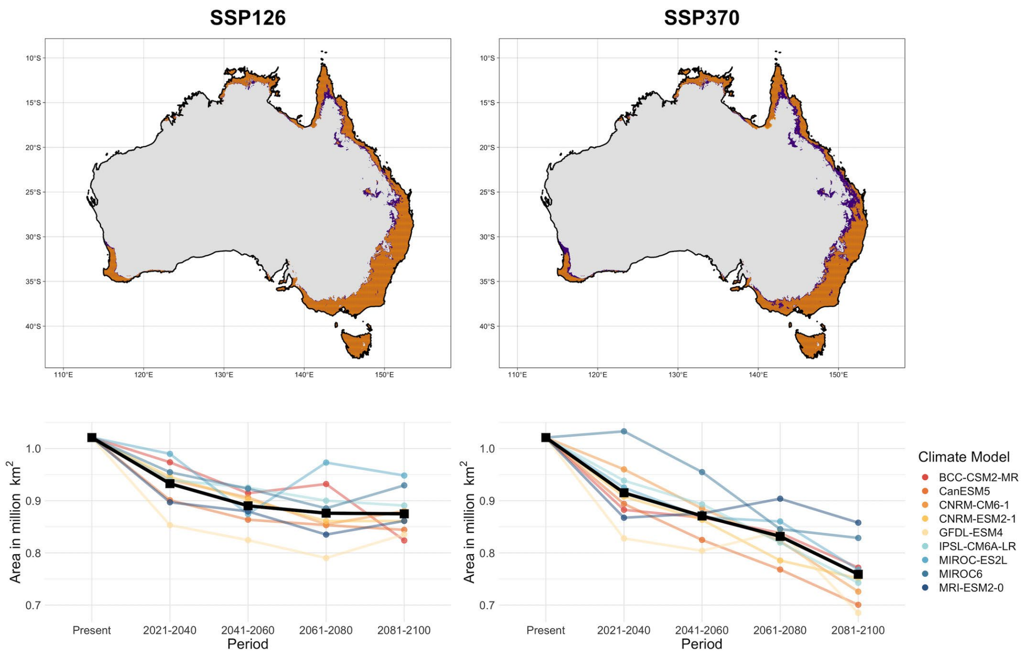


González-Orozco, C. et al. 2016. Nature Climate Change

Applications

- Evolutionary Biology
- Conservation
- Invasive Species Management
- Climate Change Impacts
- ▸ **Disease Management**

Predicting the spread of *Batrachochytrium dendrobatidis (Chytrid fungus)*



Sopniewski et al. 2022. Diversity and Distributions

This module…

- Estimating species distributions is a *data science* problem
- **Key types of data**
    - Points
    - Polygons
    - Rasters
- **Key data issues**
    - Limited data for many species
    - Biases in spatial data
    - Modelling assumptions

Spatial Data
Point Observations

Observations

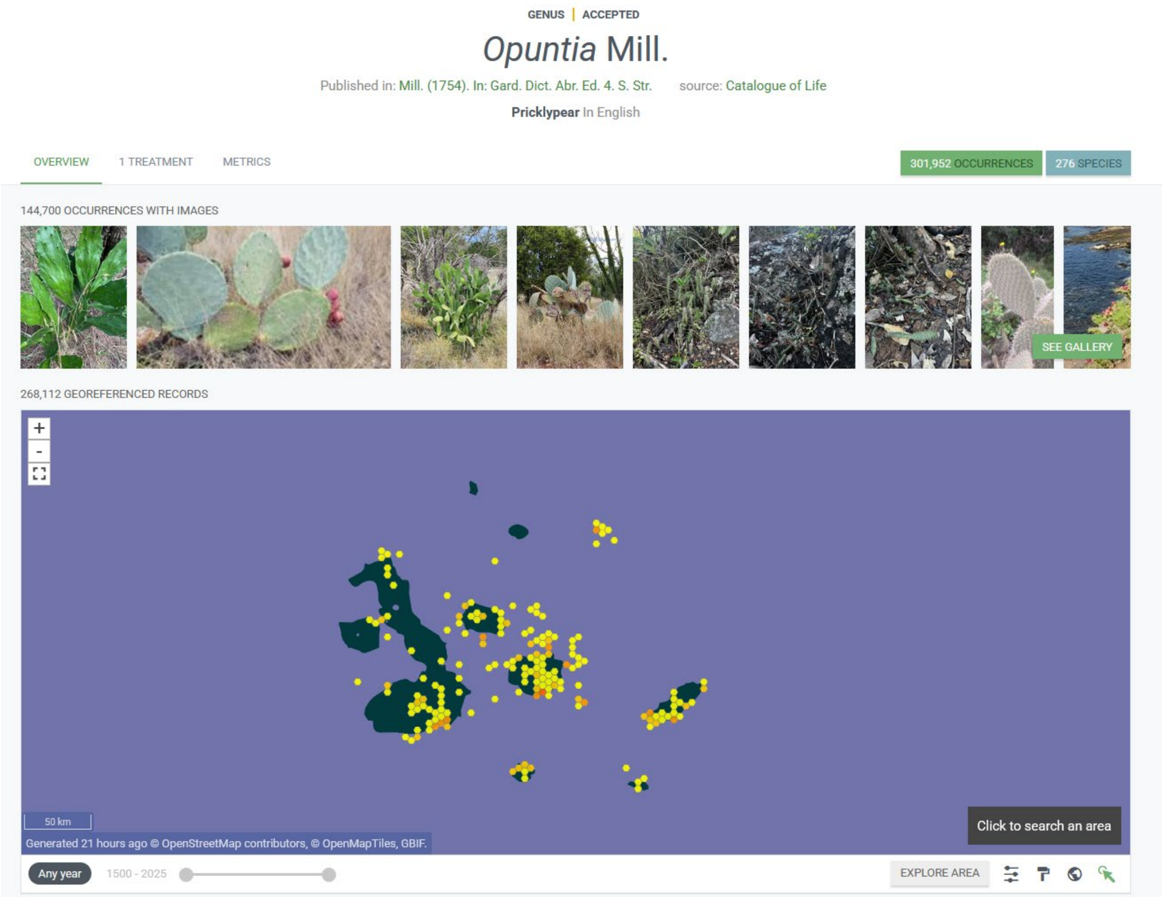
- Understanding where species are, and how this changes through time is critical to all these fields of biology
- Most of what we know about species distributions comes for natural observations

Opuntia Cacti specimens collected by Darwin in the Galapagos (Keating 2024)

Observations

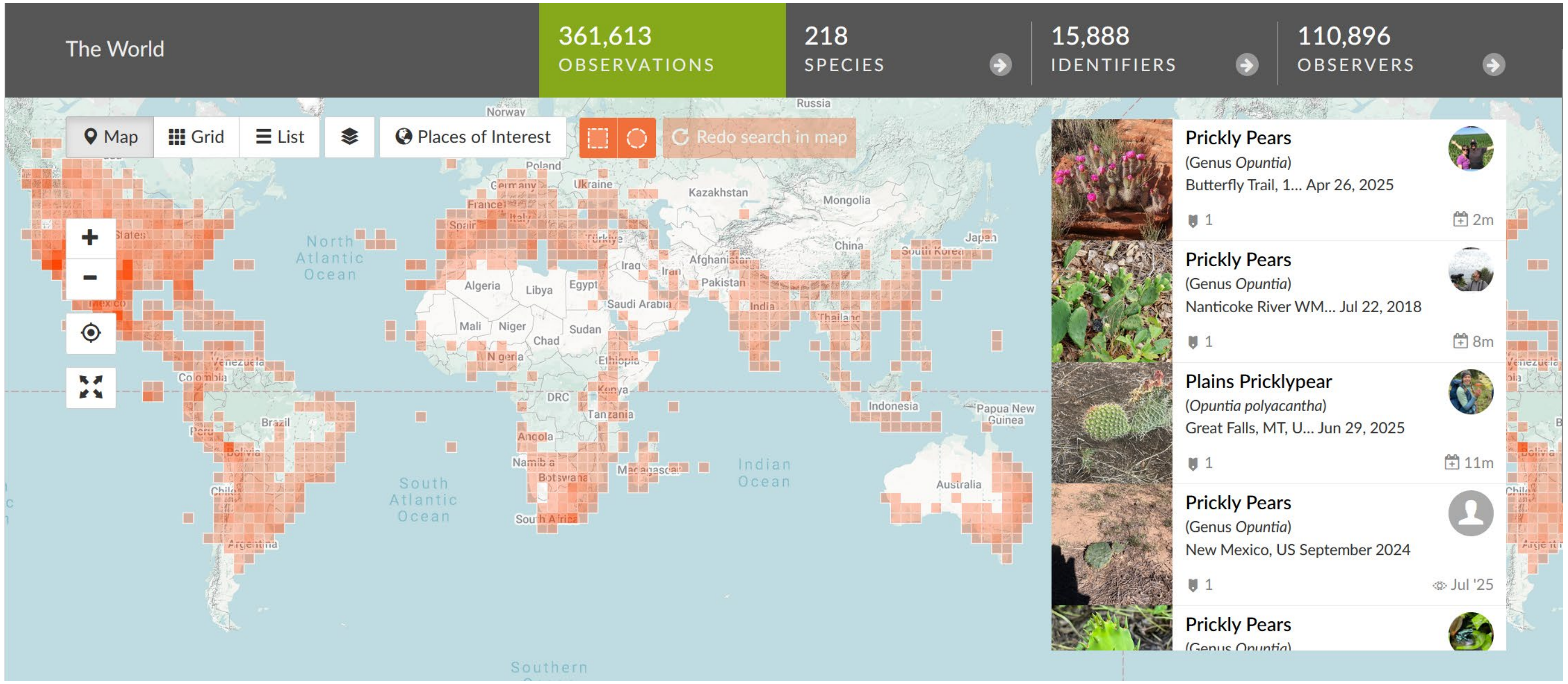
- Has **3,130,298,241** observation records as of July 2025

Global Biodiversity Information Facility



Observations

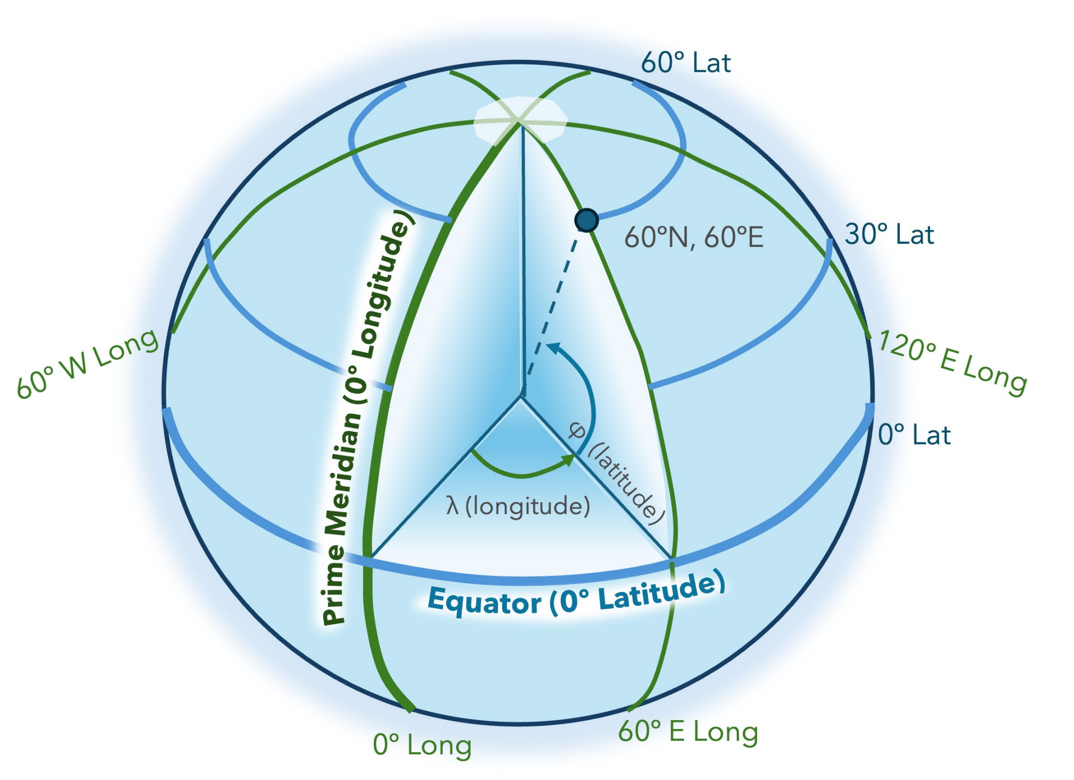
- Has **255, 762, 759** observation records from over **3 million** citizen scientists as of July 2025



Observations

- Observations are a kind of **point** data – they are coordinates in geographic space
- Defined by *x* and *y* values for a specific **coordinate reference system (CRS)**
    - *A CRS is a system for pinpointing locations on Earth using numbers*
- When the CRS is **geodetic** x=longitude and y=latitude
    - *Geodetic = based on the curved shape of the Earth*
    - World Geodetic System 84

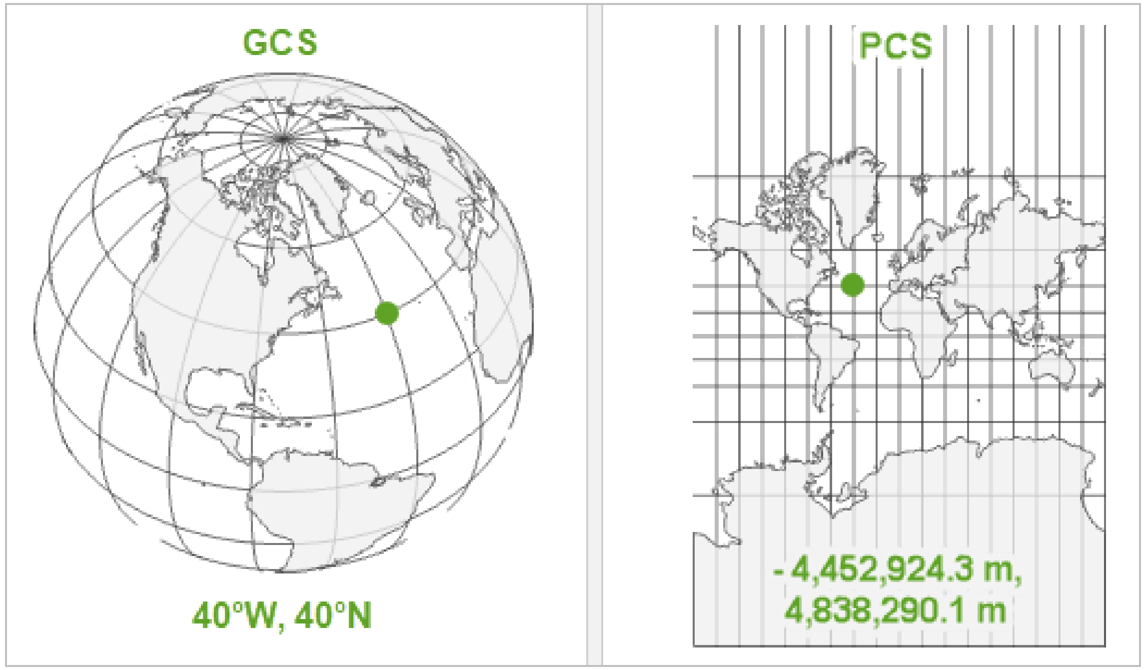
	A	B	C
1	**species**	**decimalLongitude**	**decimalLatitude**
2	Adenanthos_glabrescens	121.173677	-33.044828
3	Adenanthos_glabrescens	119.811667	-33.331389
4	Adenanthos_glabrescens	119.95	-33.383333
5	Adenanthos_glabrescens	120.367528	-33.804778
6	Adenanthos_glabrescens	120.115278	-33.294444
7	Adenanthos_glabrescens	120.07479	-33.315763
8	Adenanthos_glabrescens	119.645556	-33.490278



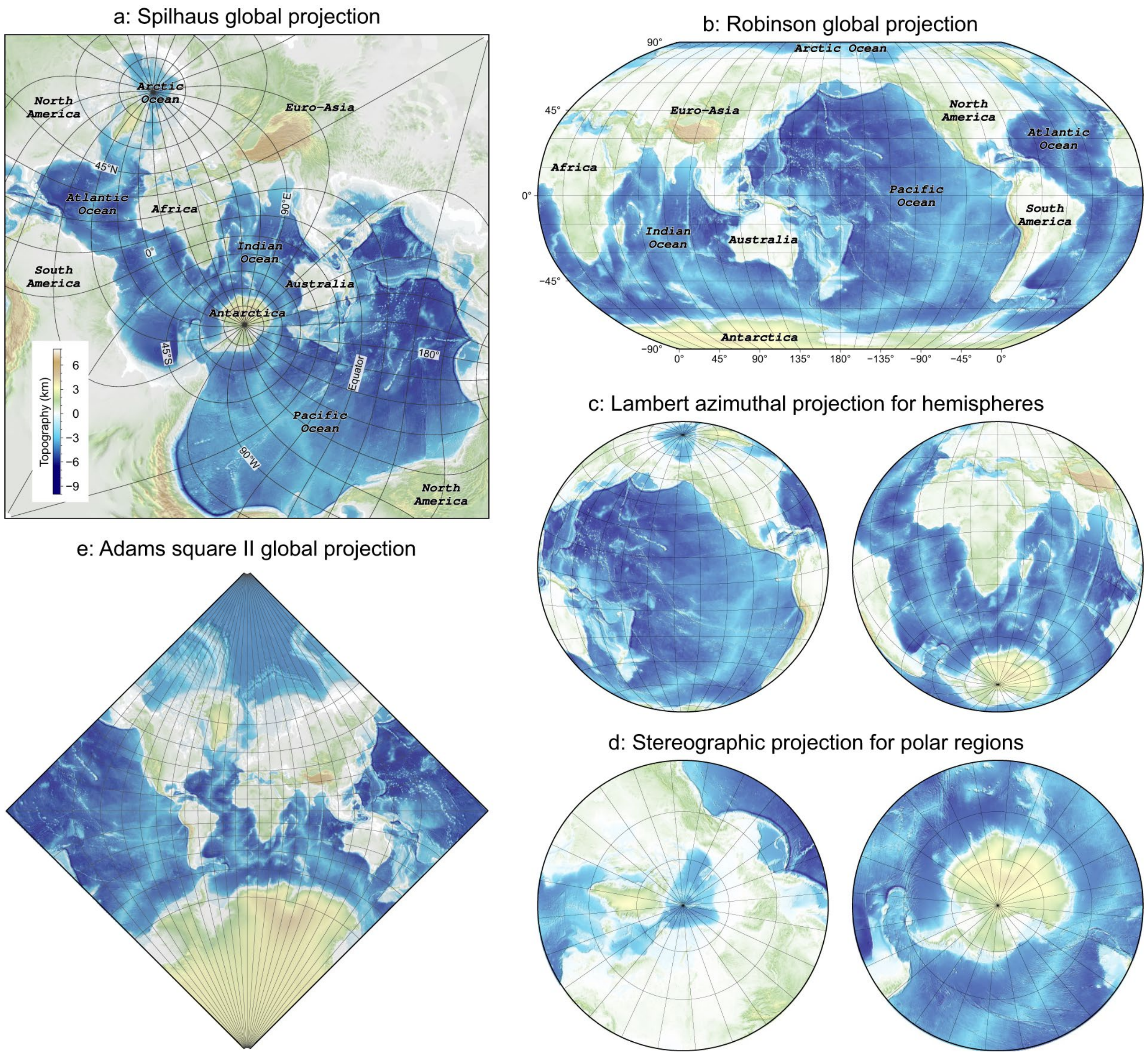
Carubio, L. F. (2025) ESRI

Observations

- Spherical Earth is 3D and Geodetic systems maintain that 3D structure
- Most maps are 2D projections
- Projecting 3D to 2D requires modifying the coordinate reference system (CRS) and often the units
  - For example, degrees to meters
- This can cause distortions in landmass



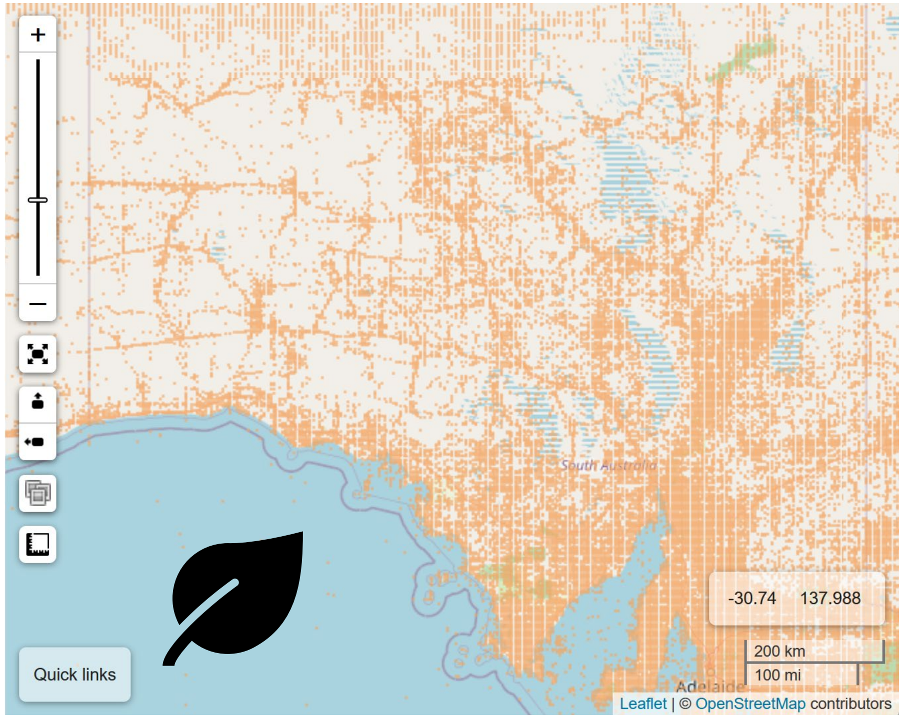
Smith, H. (2020). ESRI

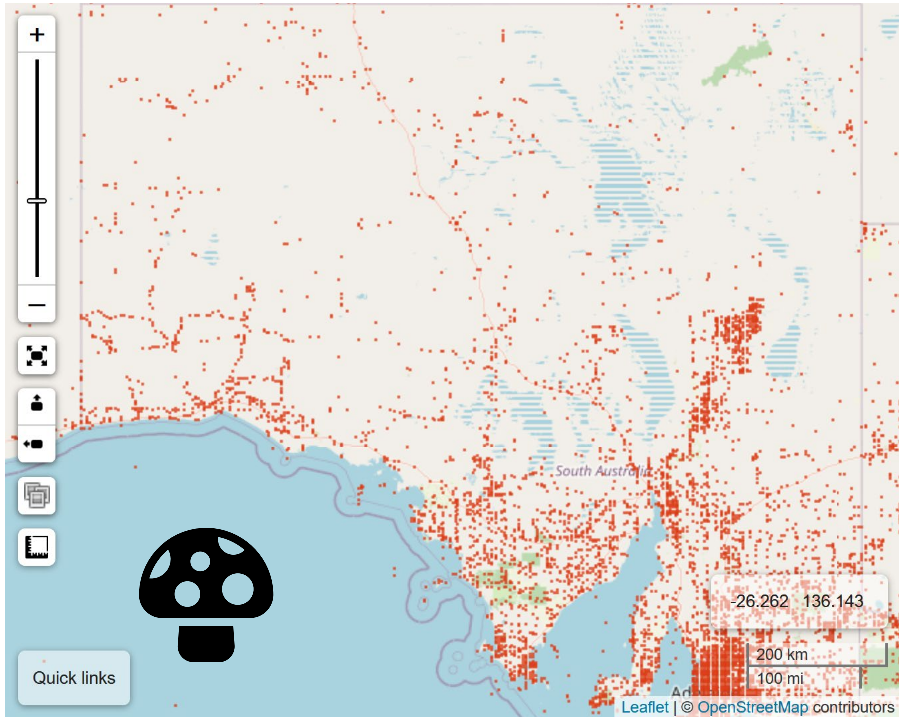


Chen et al. 2023

Observation Bias

- Observation data is **heavily biased**
- Densely populated areas are more densely sampled
- Remote areas often only sampled close to roads
- Uneven sampling across taxa





Observation Bias

Knowledge Gaps in Biological Data

Shortfall	Aspect of biodiversity	Definition
Linnean	Species	Most of the species on Earth have not been described and catalogued
Wallacean	**Geographic distribution**	**Knowledge about the geographic distribution of most species is incomplete**
Prestonian	Populations	Data on species abundance and population dynamics in space and time are often scarce
Darwinian	Evolution	Lack of knowledge about the tree of life and the evolution of species and their traits
Raunkiæran	Functional traits and ecological functions	Lack of knowledge about species' traits and their ecological functions
Hutchinsonian	Abiotic tolerances	Lack of knowledge about the responses and tolerances of species to abiotic conditions
Eltonian	Ecological interactions	Lack of knowledge on species' interactions and these interactions' effects on individual survival and fitness

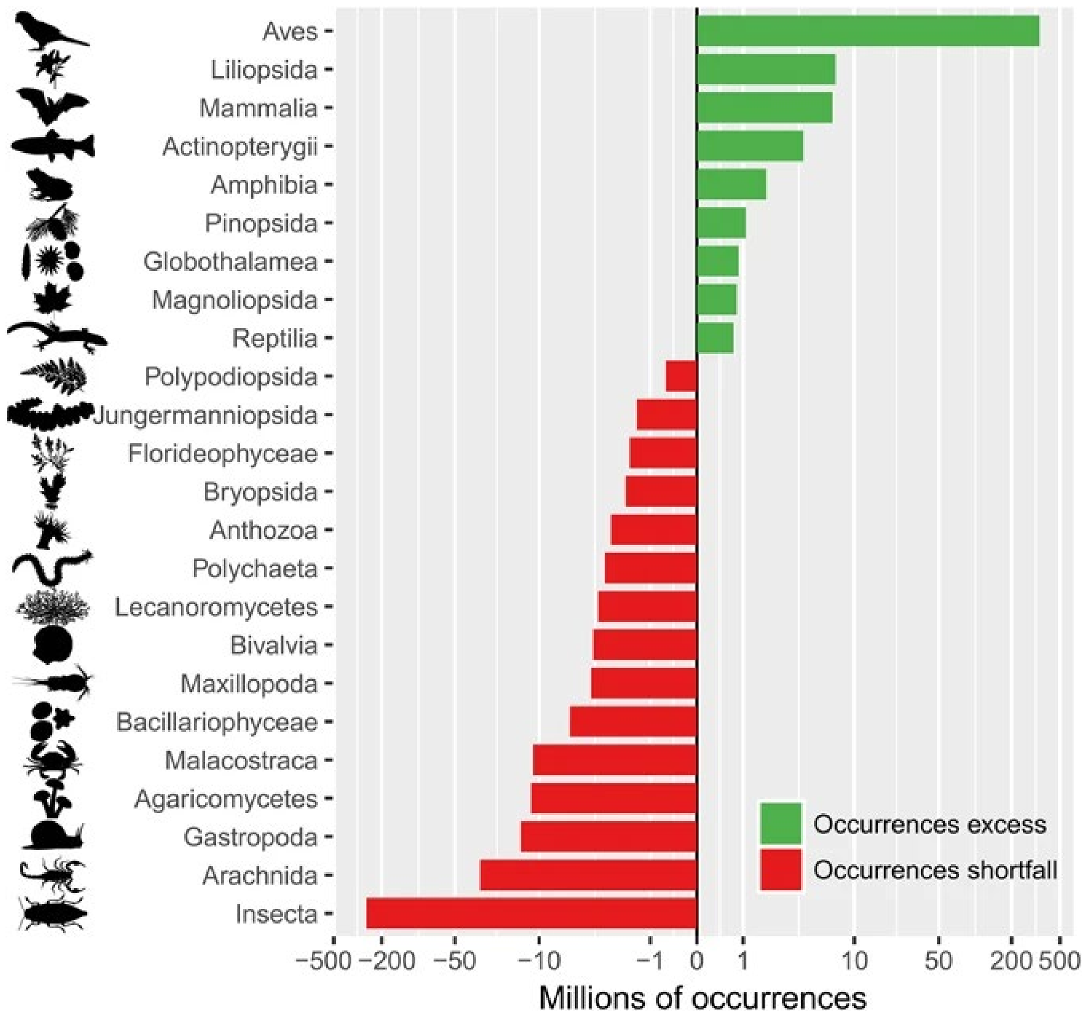
Hortal et al. 2015

Wallacean Shortfall

- *Named after Alfred Russell Wallace – the gap in our knowledge of where species live*

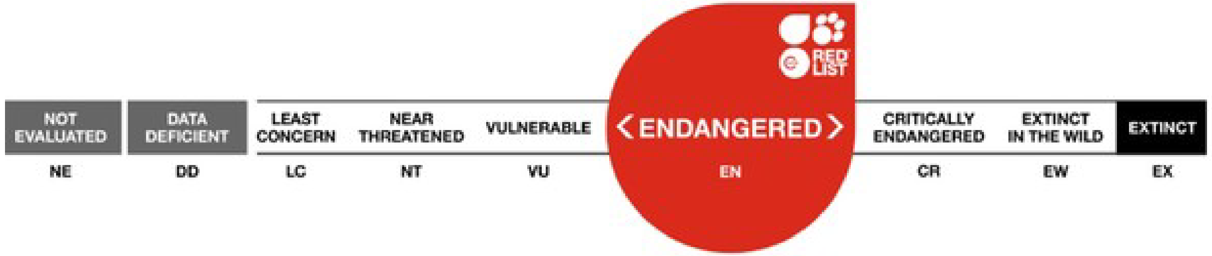
- We don’t know the geographic distribution of most species
- We know far more about the ranges of birds than insects
- >200,000 species in GBIF are known from a single record and most species have <20 records

Trudet et al. 2017

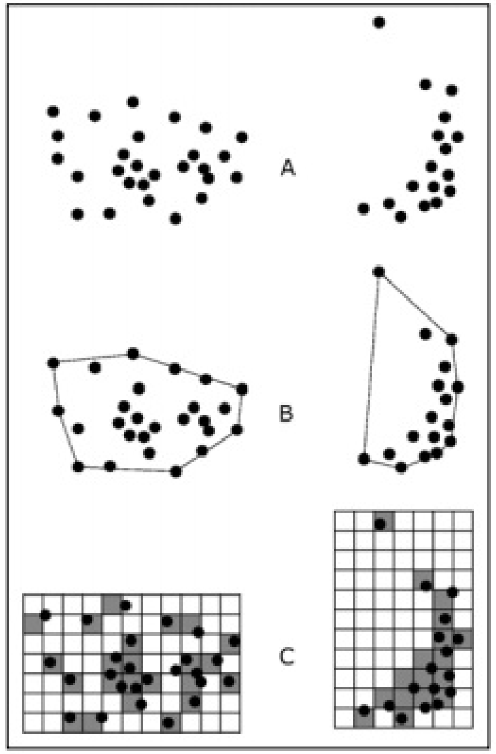


Spatial Data
Polygons

Moving from Observations to Distributions



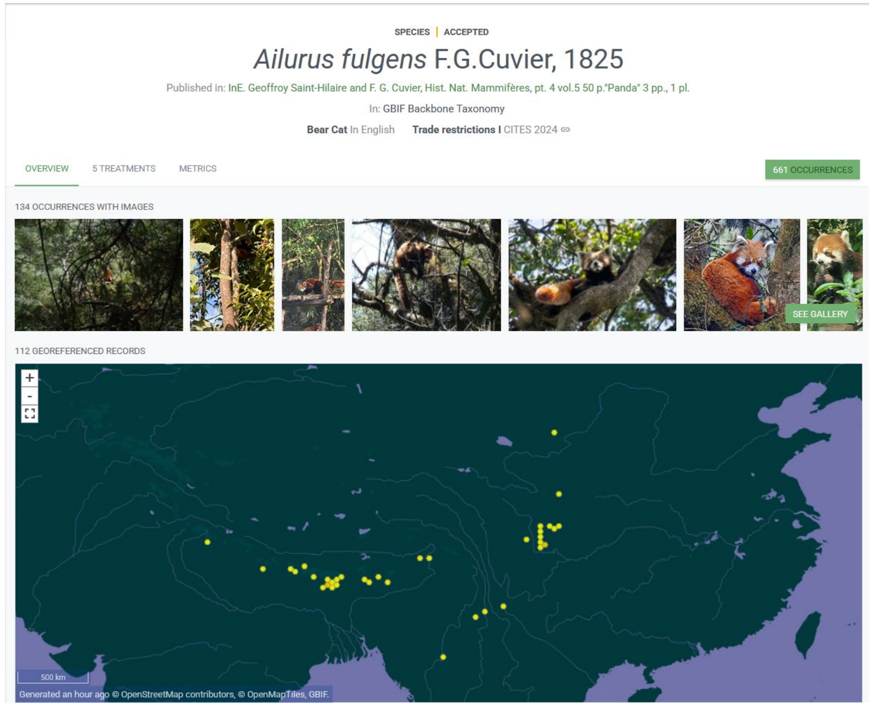
- Moving from observations to estimate of distribution
    - **Extent of Occurrence (EOO)** – *the total area where a species could possibly be found*
    - **Area of Occupancy (AOO)** – *the specific area a species actually occupies*
- AOO converts the area into a grid and counts cells with observations
- Result depends on grid cell size (spatial resolution)
- Changes in AOO and EOO are used by IUCN to assess extinction risk



IUCN

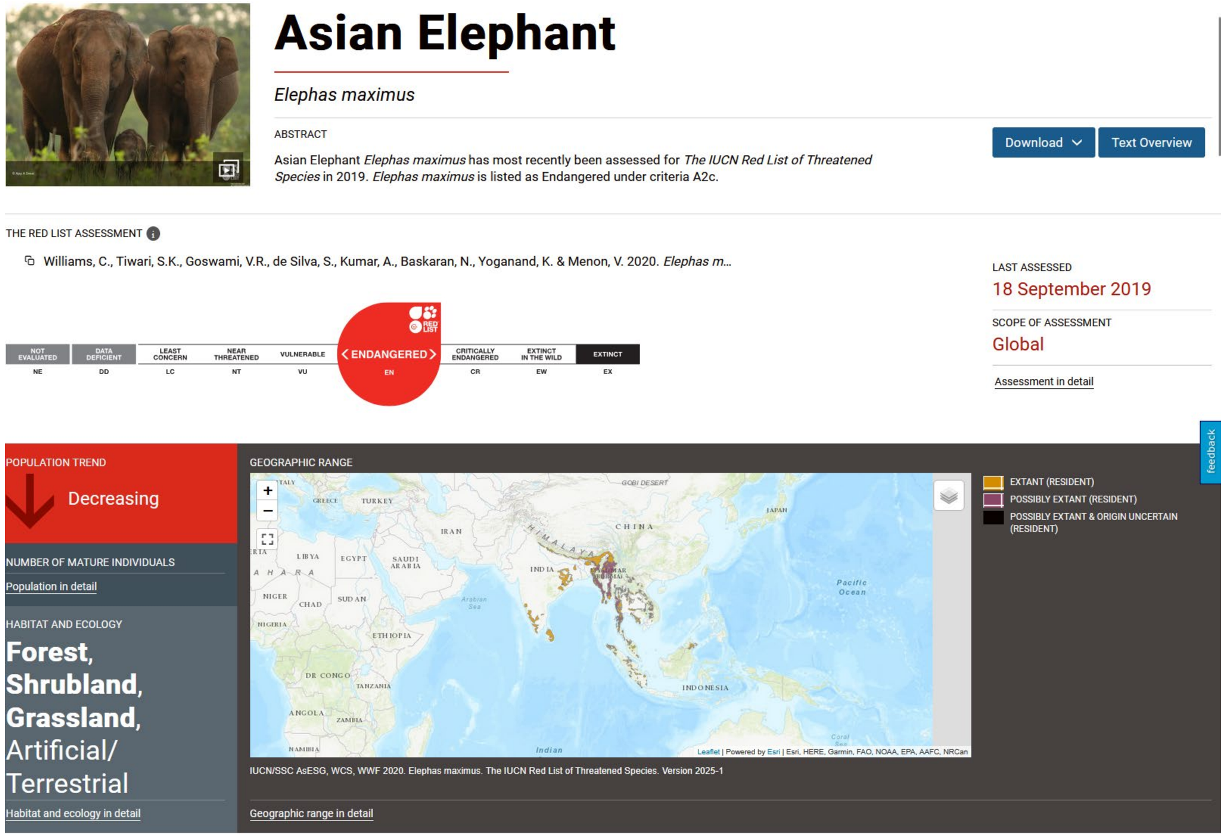
Polygons

- Observations are patchy, yet we might know the broad extent that species occur in
- Polygons are drawn, or modelled, outlines of a species range
- Can be from expert opinion or estimated from other data



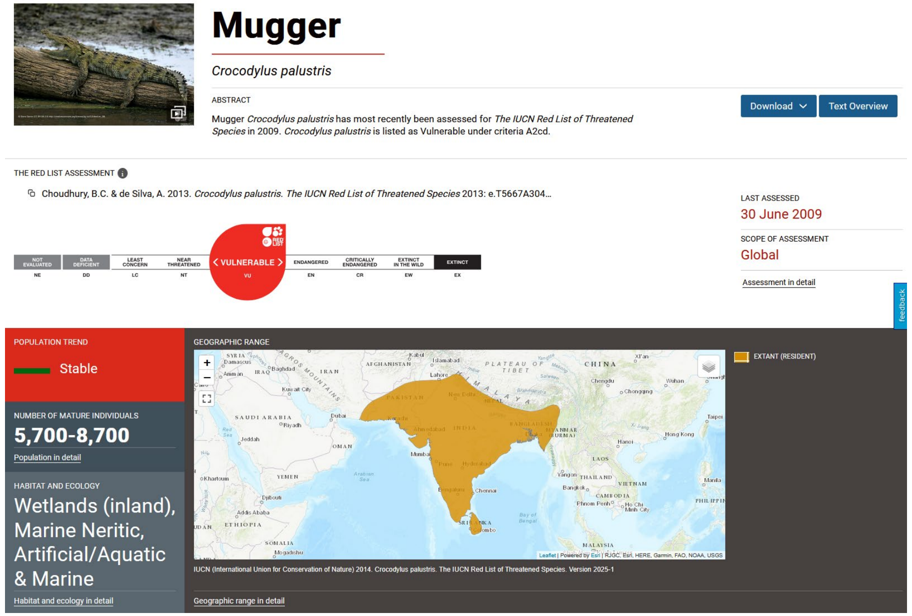
Polygons

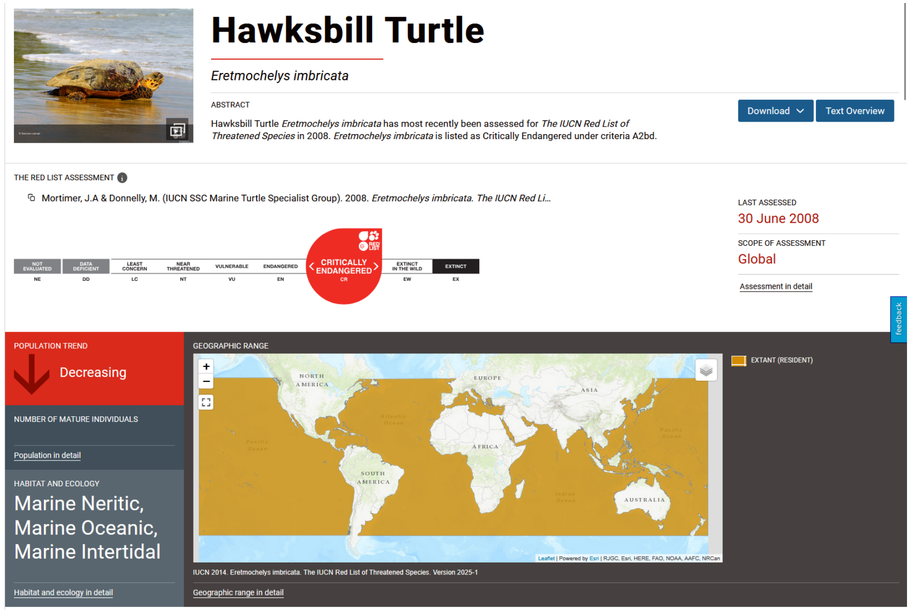
- **IUCN**: International Union for the Conservation of Nature
- Determines threat status of species
- 47,000 species are threatened with extinction
- Has assessed **169,000** species as of July 2025
- Expert range maps drawn for most species



Polygons Biases

- Often can overestimate ranges when these are uncertain.
  - Especially for migratory taxa
- Can contain areas of unsuitable habitat for some taxa





Polygons

- Polygons also represent other non-biological spatial features
  - E.g., World Database of Protected Areas (WDPA)
- Common GIS operations tell us about spatial relationships
  - **Intersection** – Do two features overlap?
  - **Containment** – Is one feature entirely within another?
  - **Proximity** – How close are two features?
  - **Adjacent** – Do two features share a border?

Species Distribution Models

Buffers and Hulls

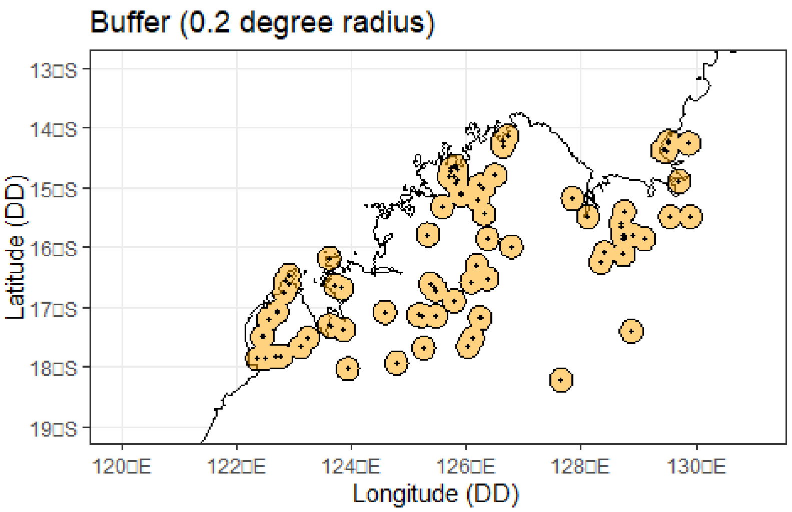
Estimating Range Polygons from Points

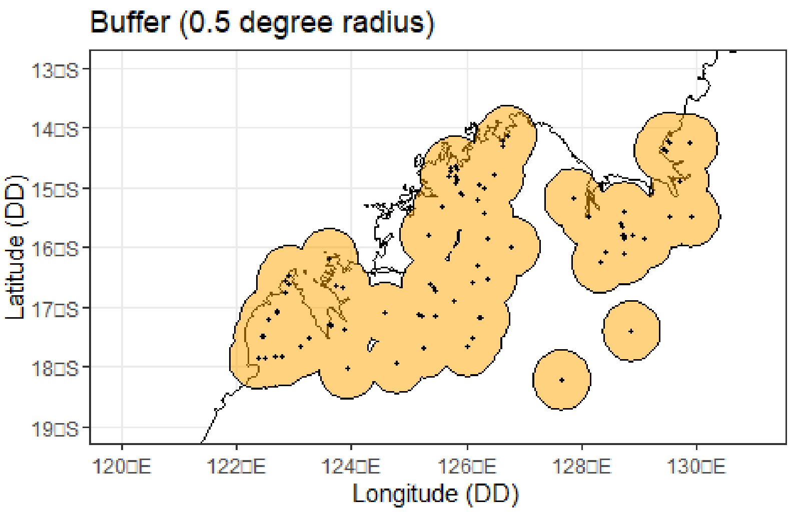
- When there are no expert polygons, one can estimate a polygon from point data
- I’ll go through three commonly used approaches:
    - **Buffer**
    - **Convex Hull**
    - **Alpha-Hull**
- I’ll call these “geometric SDMs” because they are a model of species distributions based on geometry rather than a statistical model

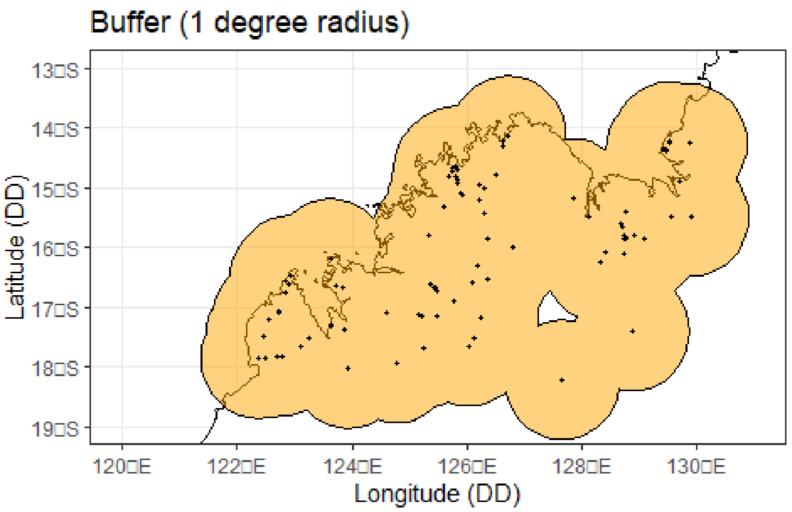
Buffer

- **Buffer:** Drawing a circle around each point at a fixed radius. Accounts for fact that if a species is observed nearby it likely is found nearby.
- May underestimate or overestimate parts of the range depending on radius

Think of it as: “draw circles around each sighting”



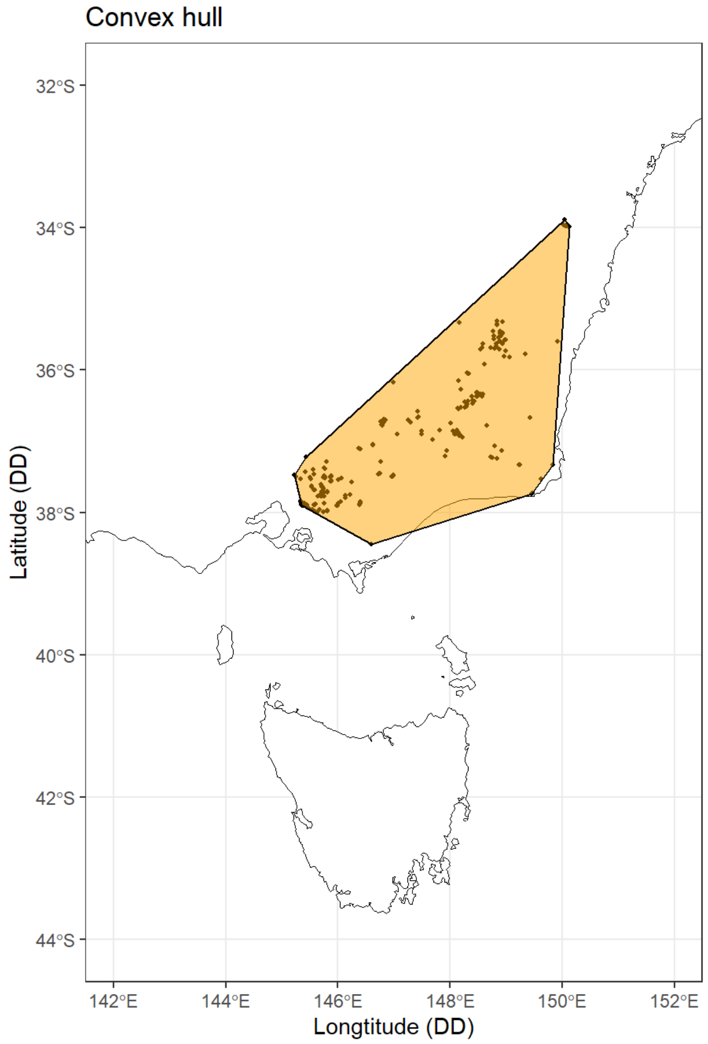




Convex Hull

- **Convex Hull:** drawing a minimum shape that encompasses all points without any internal angles >180 degrees
- May overestimate parts of the range

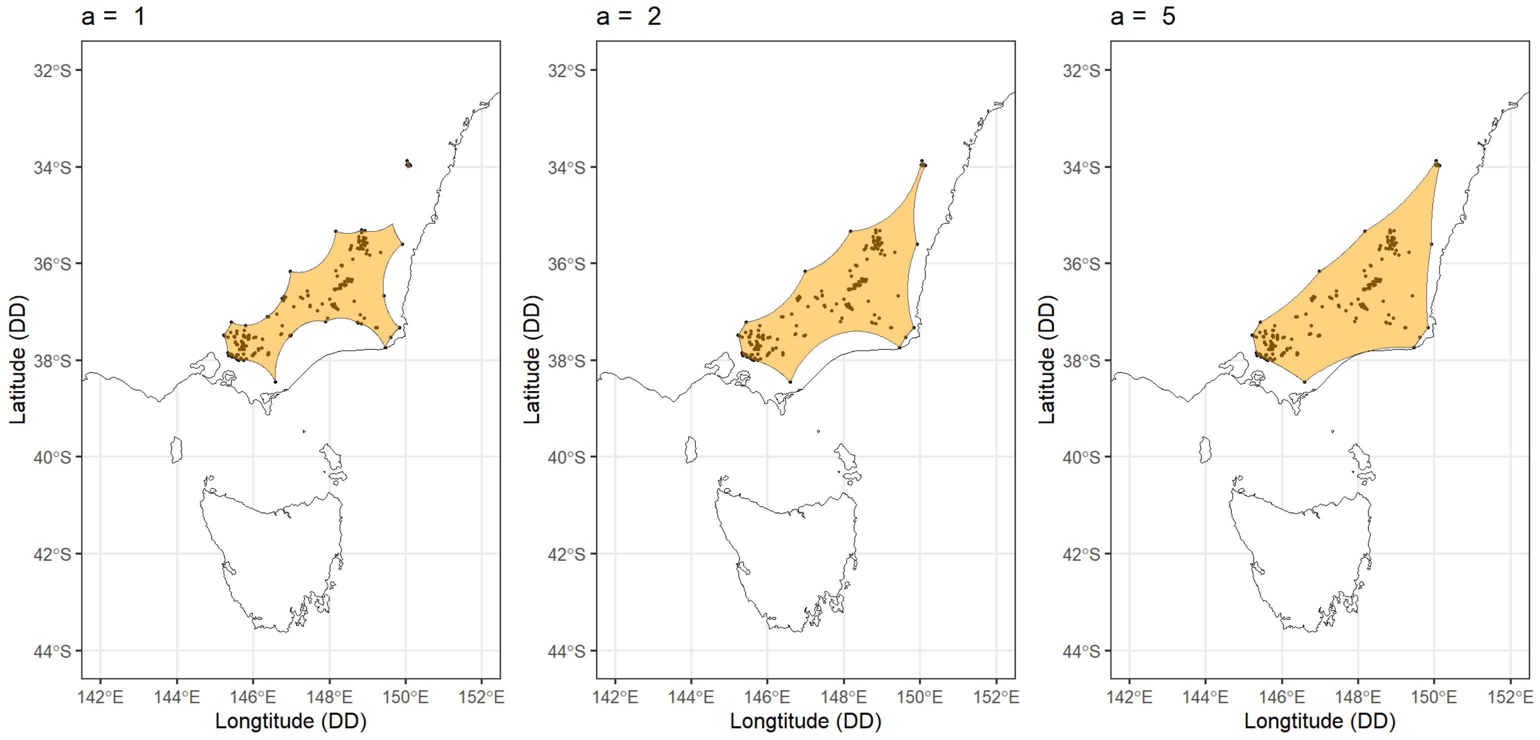
Think of it as: "stretch a rubber band around all the points"



Alpha Hull

Alpha-Hull: draws minimum shape allowing concave lines to connect points based on alpha-value

Think of it as: “like a convex hull, but the boundary can curve inward around gaps”



Geometric SDMs

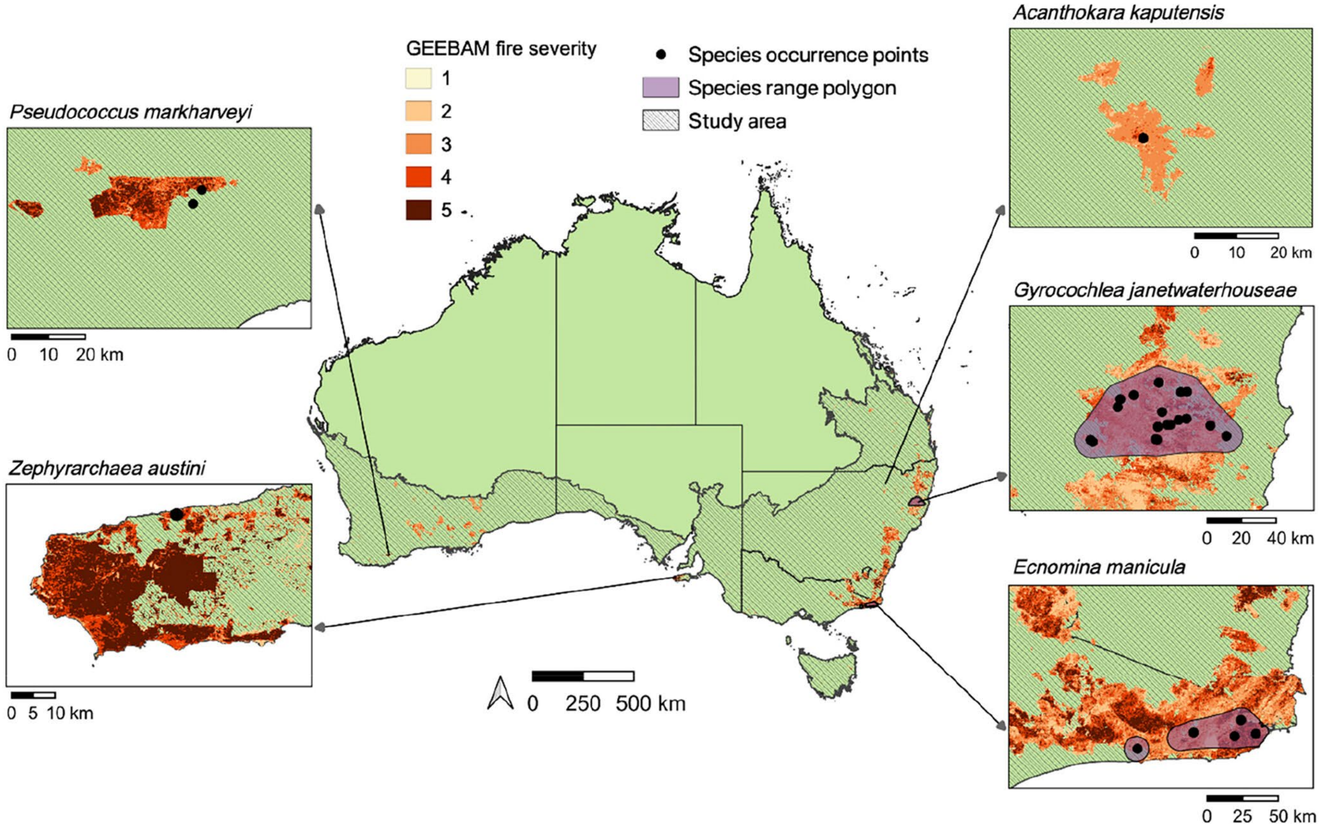
- Marsh et al. asked how much invertebrate ranges overlapped mega-fires in southeast Australia in 2019-2020
- Polygons estimated from occurrence records using alpha-hulls for **32,163 species**

RESEARCH ARTICLE

Global Ecology and Biogeography WILEY

Accounting for the neglected: Invertebrate species and the 2019–2020 Australian megafires

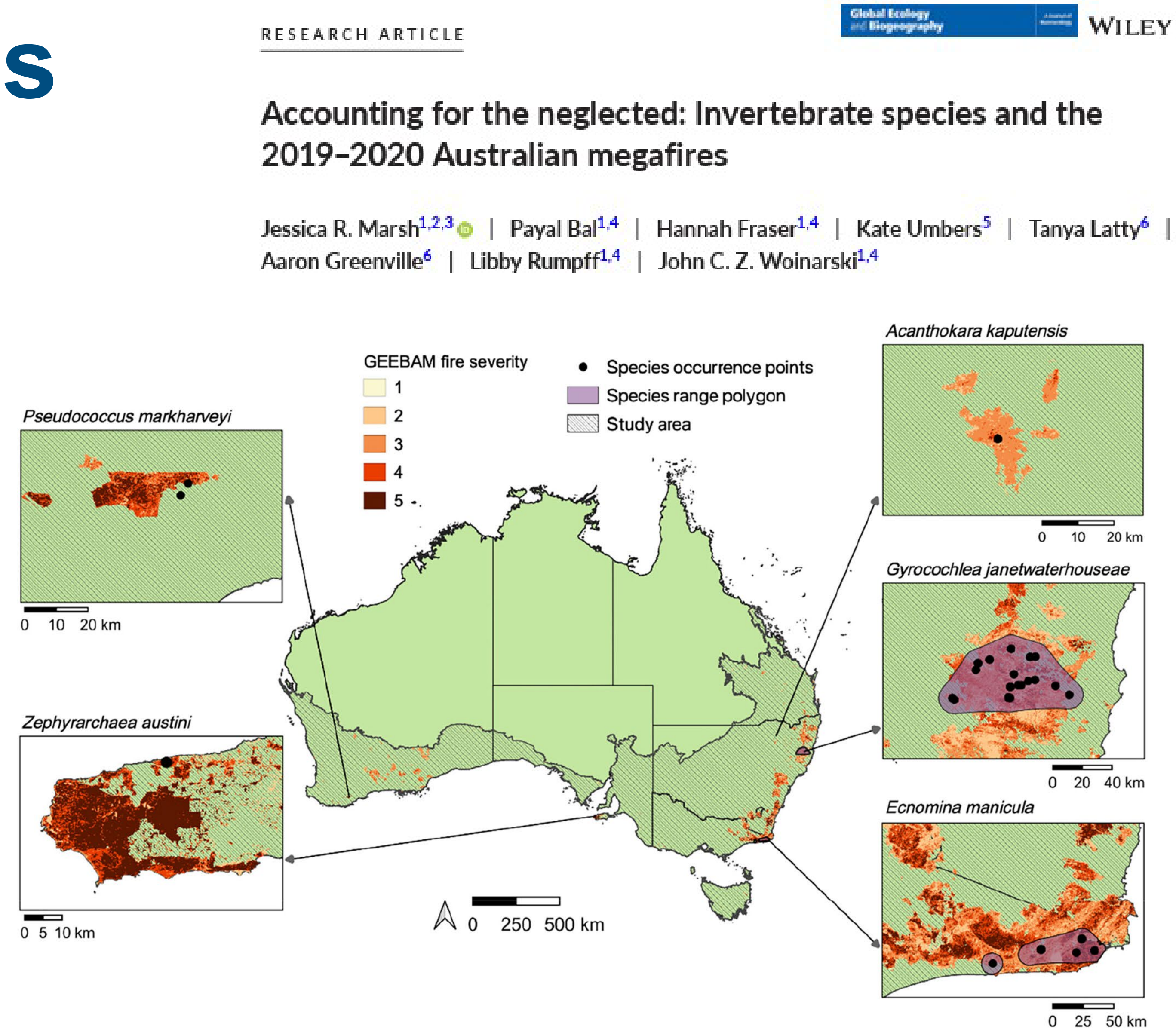
Jessica R. Marsh[1,2,3] | Payal Bal[1,4] | Hannah Fraser[1,4] | Kate Umbers[5] | Tanya Latty[6] | Aaron Greenville[6] | Libby Rumpff[1,4] | John C. Z. Woinarski[1,4]



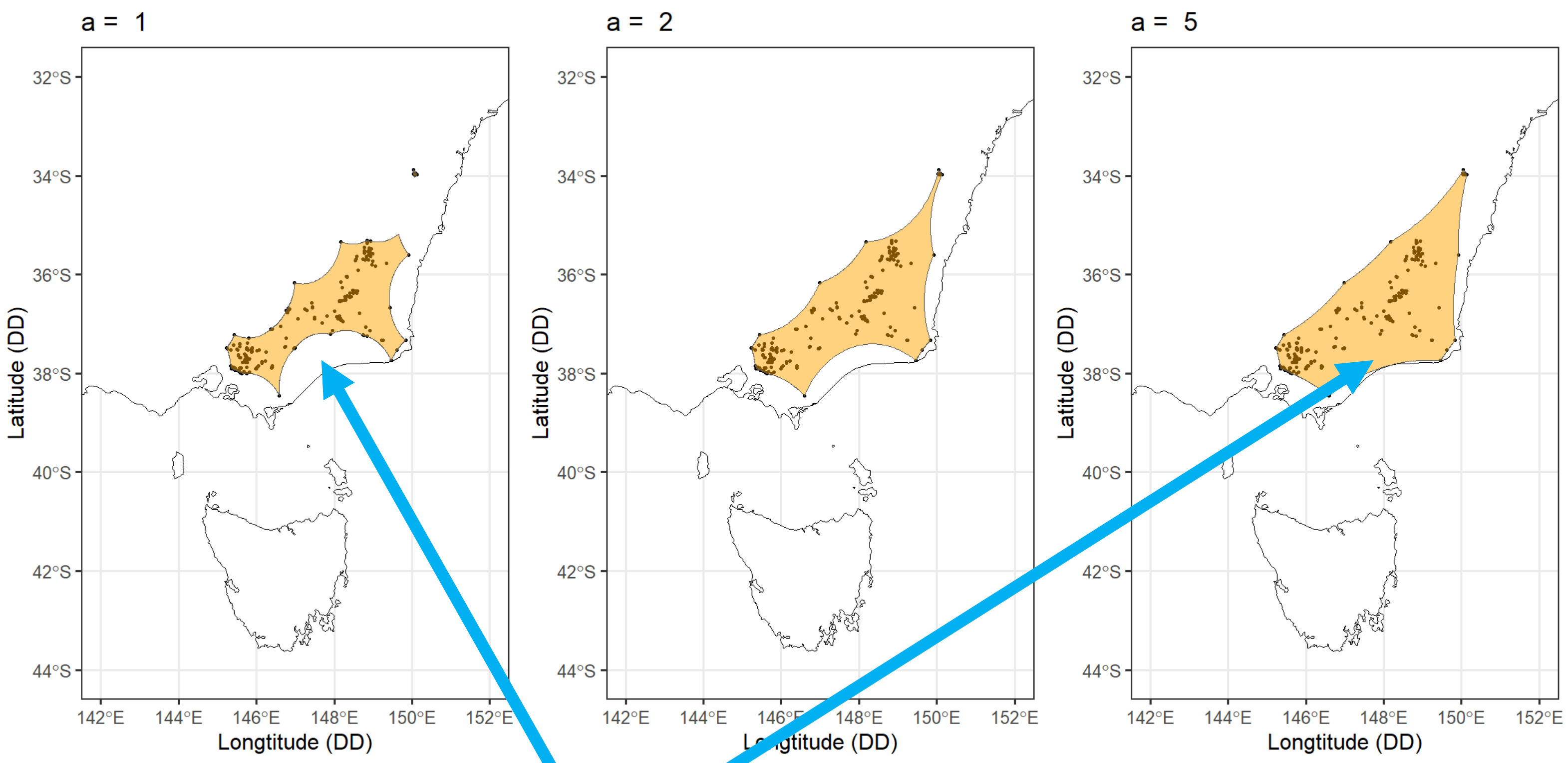
Geometric SDMs

- **13,581** invertebrate taxa had part of their range burnt
  - **382** taxa had the whole of their known range burnt
  - Further **405** taxa had 50– 99.9% of their range burnt

Kangaroo Island Assassin Spider



Limitations of Geometric SDMs

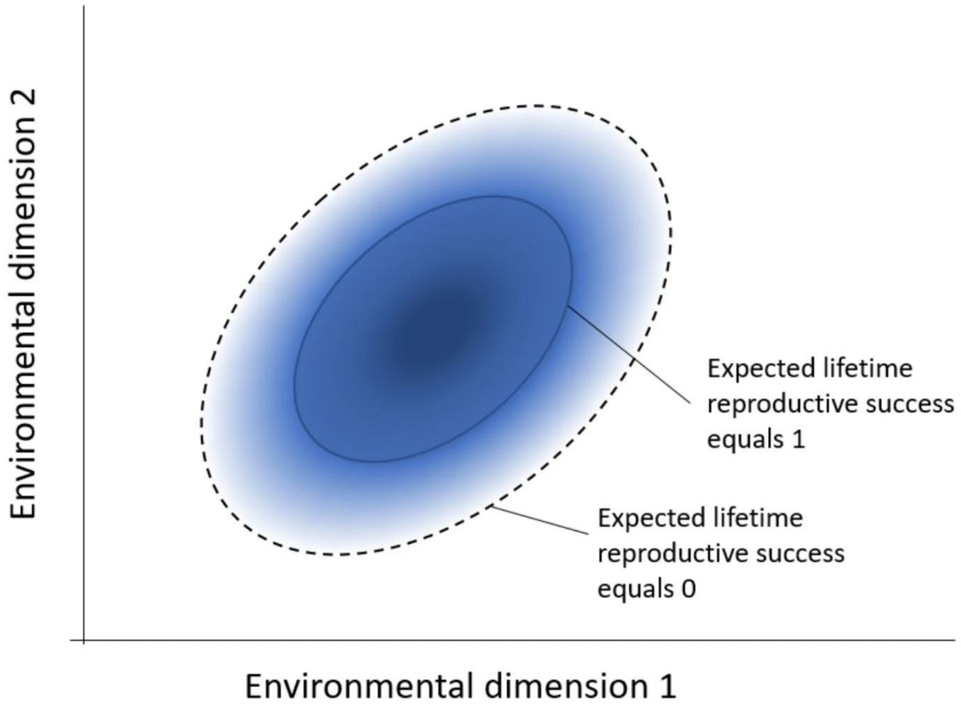


Is this species found here?
Genuinely absent? Or bias in data collection?

Limitations of Geometric SDMs

- Species can only persist or thrive in certain environments
  - n-dimensional hypervolume (Hutchinson 1978)
  - *the set of all environmental conditions (temperature, rainfall, etc.) where a species can survive, imagined as a shape in multi-dimensional space*

- Estimating a species niche may help us estimate its distributions



Takola & Schielzeth. 2022. Biol. Philos.

Take Homes

- Observations form the fundamental unit in species distribution models
- These are highly biased across space and taxa
- Estimating distributions from occurrence records involves trade-offs and assumptions
- Simple methods use geometric functions like buffers or hulls
- These may overestimate or underestimate species ranges
  - …but sometimes they're the best method available

Readings and Workshop Prep

(1) Elith and Leathwick 2009

Review paper

(2) Atlas of Living Australia

Online Lab

(3) Marsh et al 2022

Applied paper

(4) Aronsson et al. 2024

Methods paper

1. Please try to have all R packages installed before the workshop
2. Troubleshoot with Google/ChatGPT/Claude

References

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